UNIVERSITY OF UYO
DEPARTMENT OF MATHEMATICS
FIRST SEMESTER EXAMINATION - 2024/2025 SESSION

MTH 418: NUMERICAL ANALYSIS

INSTRUCTION: ANSWER ANY FOUR QUESTIONS. TIME: *180* minutes.

1(a) Show that the Chebyshev polynomials $T_n(x)$ are orthogonal on the interval $(-1,1)$ with respect to the weight function $\omega = 1/\sqrt{1-x^2}$.

(b) Using Euler's method, find an approximation for $y(1.0)$ for $-y = -\frac{1}{x}\frac{dy}{dx} + \frac{1}{x}$, $y(0)=1$ with step size $h=0.2$. Find the exact solution and compute the error for each approximation.

2(a) The improved Euler's method (Heun's method) belongs to the family of predictor-corrector methods. Use this method to compute $y(1.5)$ for the initial value problem $y'=1-xy$, $y(1)=1$ using $h=0.1$. Find the exact solution and compute the error for each approximation.

(b) In numerical computations, there are mainly two types of errors. Mention them and illustratively outline their cause(s) and how they can be reduced.

3(a) Picard's method of successive approximation is derived from transforming a first-order initial value problem $y'=f(x,y)$, $y(x_0)=y_0$ into an equivalent integral equation which can then be solved iteratively by repeatedly substituting an initial guess into the integral to generate a sequence of increasingly accurate approximations. Derive this iterative formula.

(b) Use the scheme in 3(a) to compute y(0.3) for the IVP $\frac{dy}{dx} = 2 - \frac{y}{x}$, $y(1)=2$. Stop at the third successive approximation.

4(a) Write down the iterative formula for the classical fourth-order Runge-Kutta method.

(b) Find $y(0.2)$ for the solution of $y'=2(x+y)$, $y(0)=1$ using the algorithm in 4(a) with $h=0.1$. Find the exact solution and compute the error for each approximation.

5(a) Derive the recurrence formula for generating the Chebyshev polynomials of first kind of degree n, $T_n(x)$ and write down the first 5 Chebyshev polynomials.

(b) Prove that the Chebyshev polynomial of first kind of degree n, $T_n(x)$ is a solution of the differential equation

$$(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + n^2 y = 0,$$ where $T_n(x) = \cos(n\cos^{-1}x)$.

0.05, 1.05